Student's Name: ……..…….……..

Signature: ……..

STREAM:							

525/1
CHEMISTRY
Mar./Apr. 2026
$\mathbf{2\frac{3}{4}}$ ***hours***

END OF TERM I ASSESSMENT

Uganda Advanced Certificate of Education

S.5 CHEMISTRY

Paper 1

Set I

2 Hours and 45 minutes

INSTRUCTIONS TO CANDIDATES:

This paper has **two** *sections* **A** *and* **B.**

Section **A** *has two compulsory items while* **B** *has two parts* **Part I** *and* **Part II**

Each of part I and part II has two items, Answer only **one** *from each.*

Answers to Section **A** ***must*** *be written in the spaces provided and Section* **B** ***must*** *be written in the answer booklet(s) provided*

Answer ***four*** *in all.*

Where necessary use,
The periodic table provided at the ***end*** *of the paper*

FOR EXAMINER'S USE ONLY		
ITEM	**CODE**	**SCORE**
1		
2		
3/4		
5		
TOTAL		

SECTION A

Answer **all** *questions in this section*

ITEM 1

After reports of adverse health effects among patients due to use of a herbal medicinal syrup supplied by a certain herbalist, a pharmaceutical laboratory was required to verify a plant extract used in the formulation using a mass spectrometer which indicated the presence of magnesium.

The relative atomic mass of naturally occurring magnesium is 24.3. The isotopic analysis showed that magnesium consists of three isotopes: $^{24}Mg, ^{25}Mg, and\ ^{26}Mg$, as represented in the mass spectrum. Further interpretation suggests that the percentage abundance of ^{24}Mg is eight times that of ^{26}Mg.

After initial confirmation, environmental chemists extended the investigation to nearby industrial emissions. A volatile organic compound was detected and suspected to be bromochloromethane (CH_2BrCl), a substance associated with air pollution and ozone layer depletion.

Mass spectrometric analysis of the compound showed molecular ion peaks at m/z values of $128, 130,$ and 132, resulting from the natural isotopic variations of bromine and chlorine atoms in the molecule. The analytical department requires assistance in interpreting the results. You have been approached for assistance.

Task

Help the analytical team;

(a) Explain why mass spectrometry is a suitable technique for detecting and confirming elemental composition in both medicinal plant extracts and environmental samples.

_____

_____

_____

_____

_____

(b) Explain how a mass spectrometer works to analyze a sample.(diagram not necessary)

_____

_____

_____

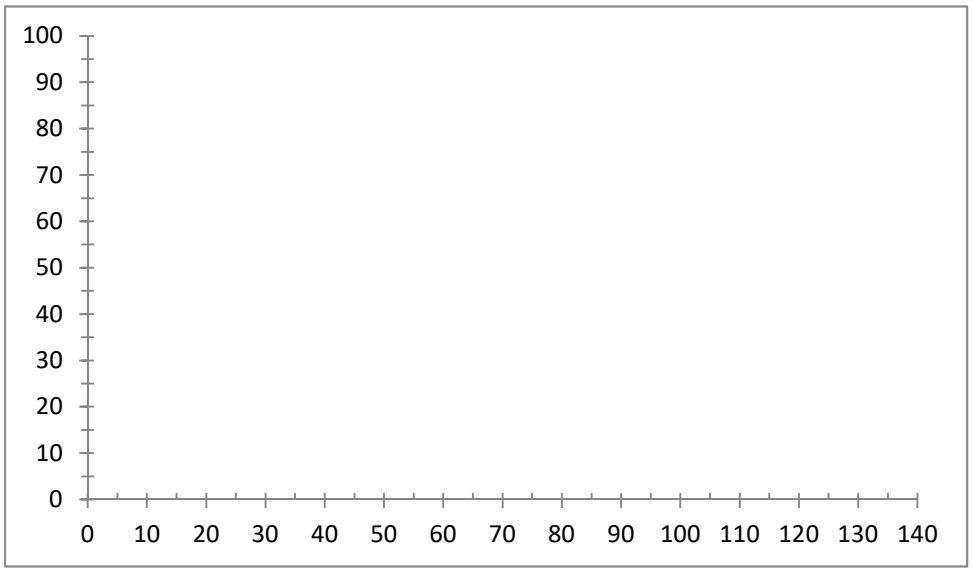
_____

_____

(c) Using the magnesium data, calculate the percentage abundance of each isotope in the plant extract.

(d) (i) Determine the formulae of the ions corresponding to the peaks at m/z 128, 130, and 132 for bromochloromethane and at least four possible fragment ions. Hence state most abundant molecular ion and fragment ion peak.
(*Given bromine exists as* $^{79}Br : ^{81}Br \approx 50.69\% : 49.31\%$, *and chlorine exists as* $^{35}Cl : ^{37}Cl \approx 75.76\% : 24.24\%$)

(ii) Sketch a mass spectrum for bromochloromethane showing the molecular ion peaks at m/z 128, 130, and 132 and at least four possible fragment ions.



2. During a marathon training session, an athlete consumes an energy drink made from powdered glucose to support aerobic respiration in muscle cells. In the body, glucose reacts with oxygen during respiration to release energy needed for sustained muscle contraction.
   The equation for aerobic respiration of glucose is:

$$C_6H_{12}O_6(s) + 6O_2(g) \rightarrow 6CO_2(g) + 6H_2O(l) + Energy$$

   In one trial, the athlete consumes 25.0 g of glucose from the energy drink. At the same time, the lungs supply 2.80 moles of oxygen to the muscle tissues during the period of intense activity.

Task

As a learner of chemistry, help the analyst to;

a) Calculate the number of moles of glucose consumed by the athlete.

b) Using the balanced equation;

(i) Determine the number of moles of oxygen required to completely react with the glucose consumed.

(ii) Identify the limiting reactant in this respiration process and justify your answer using mole calculations.

(iii) Calculate the number of moles and hence the mass of carbon dioxide produced during respiration process.

c) If only 75% of the glucose actually reacts due to physiological limitations, calculate:

(i) the moles of glucose that reacted

(ii) the moles of carbon dioxide produced

d) Using mole relationships, explain why consuming excessive glucose without sufficient oxygen supply may not increase energy availability and can reduce respiration efficiency in athletes.

SECTION B

Part I

Attempt **One** *item in this section*

Item 3

During transportation, labels were accidentally removed from eight solid elements intended for production. The supplier confirmed only that the samples are consecutive elements in Period 3 of the Periodic Table, with atomic numbers 11–18, arranged in order as E1 to E8. To prevent costly production errors, the quality control laboratory measured their first ionization energies and obtained the following results:

Element	E1	E2	E3	E4	E5	E6	E7	E8
Atomic Number	11	12	13	14	15	16	17	18
First I.E ($kJ\ mol^{-1}$)	496	738	578	786	1012	1000	1251	1521

The manager explains that: One element is needed for manufacturing strong *ionic compounds*, another for *electrical conductivity applications* and another for creating an *inert atmosphere in high temperature furnaces*. Incorrect identification leads to dangerous reactions or financial loss.

As the trainee chemist, you are required to write a detailed scientific advisory report to the plant manager explaining:

- The periodic trend shown by the ionization energy data and the atomic principles behind it.
- The identity of each element (E1–E8) with full justification.
- The reasons for any irregularities in the ionization energy trend.
- What the large jump in successive ionization energies of E2 reveals about its valence electrons. (*Given* $1^{st}, 2^{nd}$, *and* 3rd *I.Es are* $738, 1450, and\ 7730\ kJ\ mol^{-1}$ *respectively.)*
- Which elements are most suitable for metallic use, ionic compound formation, negative ion formation, and inert atmosphere application with scientific justification.

Task

Write the details of the report demonstrating logical reasoning, application of periodic trends, and clear linkage between atomic structure and industrial chemical behavior to the manger.

Item 4

A steel fabrication plant in Jinja conducts weld inspections using Cobalt-60, which emits gamma rays after beta decay. The Co-60 used has a half-life of 5.27 years and an initial activity of 4.8×10^8 disintegrations per second.

For safety audits, the plant must determine the remaining activity in Co-60 sources that have been stored for 15.8 years before disposal.

New technicians must also study the electron configurations of Cobalt (Z = 27), Co^{2+}, Co^{3+}, Nickel (Ni^{2+}), to understand corrosion resistance and alloy performance.

Task

As a learner of chemistry, help the manager;

(a) Write the electronic configurations of the given ions and explain how these electronic structures influence alloy behaviour.
(b) Write the nuclear equation for the β^- decay of Co-60 and explain the significance of the associated gamma emission in radiographic testing.
(c) Determine the activity of the Co-60 source after the stated storage duration.
(d) (i) Evaluate the industrial benefits of using Co-60 in manufacturing plants.
(ii) Suggest the environmental concerns and mitigations of using Co-60 in manufacturing plants.

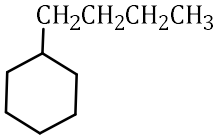
Part II

Attempt **One** *item in this section*

Item 5

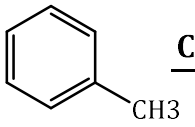
The plant has recently experienced production inefficiencies and safety concerns arising from incorrect identification of hydrocarbons and poor understanding of reaction mechanisms. Two hydrocarbon samples labeled as C_5H_{12} were found to have different boiling points and reactivities. In addition, some stored compounds were incorrectly named, and unexpected products were formed during bromination under UV light.

The plant manager has suspended further large scale processing until a clear scientific explanation is provided. You have been asked to prepare a report to clarify the structural and mechanistic issues involved. The following compounds and reactions are under review:

1. $(CH_3)_2CHCH_2CH_2CH_2CH_3$ **2.** $CH_3CH_2\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}}CH_2CH_3$ **3.** $CH_3\overset{\overset{CH_2CH_3}{|}}{\underset{\underset{CH_2CH_3}{|}}{C}}HCH_2CHCH_2CH_3$ **4.** 

Halogenation Reactions Under U.V Light:

1. $CH_3CH_2CH_3 \xrightarrow{\mathbf{Br_2 + U.V\ Light}} ??$

2. 

$\xrightarrow{\mathbf{Cl_2 + U.V\ Light}} ??$

The manager seeks to know;

- The meaning of structural isomerism, description of the three types of structural isomerism, giving one suitable example in each case and the correct IUPAC names of the compounds under review listed above.
- The meaning of the following reaction types, giving one appropriate organic example in each case: Nucleophilic addition, Nucleophilic substitution, Electrophilic substitution, Electrophilic addition, Elimination and Free radical substitution
- The type of reaction occurring, and well outlined mechanism for each of the two halogenation reactions indicating clear stages.

You have been approached by the manager for assistance.

Task

Write a brief, well structured report to the plant manager.

THE PERIODIC TABLE

1	2											3	4	5	6	7	8
1.0 H 1																1.0 H 1	4.0 He 2
6.9 Li 3	9.0 Be 4											10.8 B 5	12.0 C 6	14.0 N 7	16.0 O 8	19.0 F 9	20.2 Ne 10
23.0 Na 11	24.3 Mg 12											27.0 Al 13	28.1 Si 14	31.0 P 15	32.1 S 16	35.4 Cl 17	40.0 Ar 18
39.1 K 19	40.1 Ca 20	45.0 Sc 21	47.9 Ti 22	50.9 V 23	52.0 Cr 24	54.9 Mn 25	55.8 Fe 26	58.9 Co 27	58.7 Ni 28	63.5 Cu 29	65.7 Zn 30	69.7 Ga 31	72.6 Ge 32	74.9 As 33	79.0 Se 34	79.9 Br 35	83.8 Kr 36
85.5 Rb 37	87.6 Sr 38	88.9 Y 39	91.2 Zr 40	92.9 Nb 41	95.9 Mo 42	98.9 Te 43	101 Ru 44	103 Rh 45	106 Pd 46	108 Ag 47	112 Cd 48	115 In 49	119 Sn 50	122 Sb 51	128 Te 52	127 I 53	131 Xe 54
133 Cs 55	137 Ba 56	139 La 57	178 Hf 72	181 Ta 73	184 W 74	186 Re 75	190 Os 76	192 Ir 77	195 Pt 78	197 Au 79	201 Hg 80	204 Ti 81	207 Pb 82	209 Bi 83	209 Po 84	210 At 85	222 Rn 86
223 Fr 87	226 Ra 88	227 Ac 89															
			139 La 57	140 Ce 58	141 Pr 59	144 Nd 60	147 Pm 61	150 Sm 62	152 Eu 63	157 Gd 64	159 Tb 65	162 Dy 66	165 Ho 67	167 Er 68	169 Tm 68	173 Yb 70	175 Lu 71
			227 Ac 89	232 Th 90	231 Pa 91	238 U 92	237 Np 93	244 Pu 94	243 Am 95	247 Cm 96	247 Bk 97	251 Cf 98	254 Es 99	257 Fm 100	256 Md 101	254 No 102	260 Lw 103

END